IOT Based System for Detecting Accidents and Alerting Nearby Hospitals

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***Abstract*: Utilizing IoT technology, this innovative system revolutionizes the response to road accidents, prioritizing swift detection and immediate medical assistance. Embedded with accident-detection sensors like accelerometers and gyroscopes, vehicles continuously relay data to a central server. Advanced algorithms swiftly analyze this data, identifying patterns indicative of accidents and triggering alerts. A centralized database of local hospitals facilitates real-time transmission of alerts, ensuring prompt responses. Hospitals receive alerts through a dedicated web application, allowing for efficient management and allocation of resources. Should the primary hospital be unable to accommodate, the system intelligently forwards alerts to the nearest available healthcare facility. This seamless integration of IoT technology not only accelerates emergency response times but also optimizes resource allocation, potentially mitigating the severity of injuries and saving lives. By streamlining the process of accident detection and hospital alerting, this system addresses the critical need for rapid medical assistance on the road, ultimately enhancing the effectiveness of emergency medical services and improving outcomes for accident victims.**

Keywords: Accelerometers, IoT technology, Gyroscopes, Emergency medical services, Advanced algorithms.

I INTRODUCTION

In a world where road accidents persistently claim lives, there's an urgent need for innovative solutions. The critical moment of an accident demands timely medical assistance, a challenge plagued by response delays and resource allocation inefficiencies. It shines brightly in this dark scenario. It harnesses IoT's potential to link accidents with swift hospital responses. IoT's healthcare domain enables real-time accident detection, instant hospital alerts, and seamless communication via a web application. This groundbreaking solution aims to transform emergency response, ensuring accident victims receive prompt medical aid. Upon detecting an accident, vehicle-installed IoT devices swiftly alert the nearest hospital. If declined, the system smoothly redirects alerts to the next closest hospital, optimizing resource usage and potentially reducing fatalities.

The paper is organized as follows with the literature survey in section II which is followed by the proposed system in section III which is followed by methodologies used on accident prediction in section IV then results in section V and the conclusion in section VI.

II LITERATURE SURVEY

Before diving into our proposed system, it is essential to explore the wider research and technological context. This review encompasses various areas including IoT-based accident detection, emergency response systems, hospital alert mechanisms, resource allocation strategies, and healthcare web applications. By comprehensively understanding the strengths and limitations of these existing solutions, we can better position our innovative approach. Our aim is to enhance emergency response, improve resource allocation, and optimize healthcare services during accidents and emergencies. Here, we will discuss the diverse techniques and approaches employed in numerous research papers.

Adnan Bin Faiz, Ahmed Imteaj, and Mahfuzulhoq Chowdhury [1] present a study on an Android-based application employing the GPS receiver, pressure sensor, and accelerometer sensor within smartphones. This application is designed to identify sudden changes in deceleration, pressure, and tilt, suggesting an accident occurrence. Upon detecting these conditions, the application promptly sends an emergency alert message to the nearest police station and healthcare center. Additionally, the system incorporates an emergency switch option, enabling drivers to send alert messages immediately, bypassing the need for accident detection conditions to be met.

Nicky Kattukkaran, Arun George, and Mithun Haridas T.P [2] present a study on an intelligent accident detection and alert system designed to tackle the rising incidence of road accidents, particularly concerning two-wheelers. The system employs accelerometers and heartbeat sensors to identify accidents and evaluate their severity. Subsequently, it dispatches alerts to nearby medical centers and notifies friends and relatives through a mobile application. The paper details the system's architecture, hardware elements, and Bluetooth integration with mobile devices, highlighting the crucial role of swift medical aid in mitigating accident-related fatalities.

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Anil Gadekar, Shubham Patil, Atul More, and Sachin Hiray [3] present a pioneering method centered on harnessing IoT technology to develop a smart application dedicated to post-accident management. This application aims to rapidly detect accidents and promptly transmit incident details to emergency services, hospitals, and law enforcement through online messaging. Its primary aim is to shorten response times, ultimately improving the likelihood of saving lives by minimizing delays

Nagarjuna R Vatti, PrasannaLakshmi Vatti, Rambabu Vatti, and Chandrashekhar Garde [4] have introduced a model addressing the escalating concern of fatal and debilitating road accidents. This model underscores the necessity for a system capable of not only detecting accidents but also communicating with relevant stakeholders. The proposed system utilizes vibration and gyro sensors to identify accidents. Upon detection, it dispatches messages to emergency contacts, nearby hospitals, and law enforcement agencies. Additionally, it monitors the driver's heart rate and activates vehicle lights to alert passersby about the accident. The system's efficacy was evaluated through experiments employing a remote-controlled toy car.

Sayanee Nanda, Harshada Joshi, and Smita Khairnar [5] present a paper detailing an IoT-based smart system aimed at both preventing and detecting accidents. This system integrates multiple components including RFIDs, sensors, V2V communication, Raspberry Pi, and MEC architecture. By combining these elements, the system offers a unique prevention and detection mechanism ensuring safety and timely measures to prevent loss of life.

Prashant Kapri, Shubham Patne, and Arul Shalom A [6] outline a proposed system architecture comprising two pivotal components: the ADAS server and the ADAS software. The ADAS software, installed on smartphones, is tasked with accident detection and transmitting information to the server. Subsequently, the ADAS server statically analyzes the received data and notifies or contacts the client's family, friends, or emergency responders, such as ambulances, about the accident. Furthermore, the ADAS server facilitates the emergency dispatch of the nearest available ambulance to the accident location.

Wan-Jung Chang, Liang-Bi Chen, and Ke-Yu SU [7] present the proposed model of the DeepCrash system, comprising an IVI telematics platform equipped with vehicle collision detection sensors, along with a cloud-based deep learning server and management platform. The deep learning model is specifically trained by the cloud-based server, employing a widely used open deep learning framework, to identify traffic collision events accurately.

Fizzah Bhatti, Munam ali shah, Carsten Maple, and Saif Ul Islam [8] The proposed system consists of two phases: accident detection and the notification system. The system relies on the ubiquity of sensor-rich mobile phones for the detection of accidents. The proposed system uses multiple smartphone sensors including accelerometer, GPS, pressure, and microphone acquisition to detect accidents. An accident is detected through threshold analysis. Upon detecting an accident, the system informs the nearest hospitals and ambulances. The system requires Internet connectivity to function.

R. Raffik, M. Michael Jones, T. Murugajothi, and B. Kannadasan [9] present an automatic accident detection system utilizing a blend of GPS, GSM, and gyroscope sensors for real-time accident detection and reporting. This system is crafted to be both cost-efficient and efficient, offering integration with emergency services to swiftly dispatch rescue teams to accident locations.

S. Sasi Priya, Arfana Suraiba, Ajaai R, and Harini S [10] introduce a mobile tracking application tailored for emergency ambulance services, leveraging smartphone technology. This application enables users to promptly request and monitor nearby ambulances in real-time. In case of an emergency, users can book an ambulance, track its location on a map, and oversee the patient's transit to the hospital. The system utilizes web services and navigation to continuously update ambulance locations, potentially aligning with traffic signals to optimize routes.

CK Gomathy and Neelam Sirisha Reddy [11] propose an Accident Detection and Smart Rescue System utilizing IoT technology for vehicle accident location and alerting. The system integrates GPS, GSM, accelerometer, and RF server components. It comprises four units: vehicle, control, ambulance, and traffic junction. The vehicle unit locates accidents via GPS and transmits the data to the control unit via GSM. The control unit manages a hospital database, dispatching information to nearby hospitals to deploy ambulances. Additionally, the system provides real-time tracking of accident locations and sends alerts to registered mobile numbers.

Gopal Sunil Rathod, Rani Chandrabhan Tipale, and Kapil Jajulwar [12] introduce a system that merges Internet of Things (IoT) technology with machine learning to bolster traffic safety. An IoT module identifies accidents, gathers diverse data including location, pressure, force, and speed, and transmits this information to the cloud. Within the cloud, a machine learning model validates the accident data and initiates emergency responses, thereby minimizing response times and potentially safeguarding lives by alerting traffic monitors or rescue teams.

S. Ram Prasath, R. Santhana Krishnan, S. Mohanap Priya, and M. Arun Kumar [13] introduce a system designed to prevent hit-and-run cases in India through an automatic accident detection mechanism installed in vehicles. Upon detecting an accident, the system promptly disables the vehicle's engine and simultaneously informs the nearby police station with the owner's details. Additionally, it shares the accident location and vehicle information with the insurance company for precise accident analysis. This technology aims to deter hit-and-run incidents by ensuring swift accountability measures.

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Abdulkadir Shehu Bari, Muhammad Abubakar Falalu, Muhammad Auwal Umar, Yakubu Yunusa Sulaiman, Abdullahi Mansur Gamble, and Muhammad Ahmad Baballe [14] propose a system for accident detection and response that utilizes accelerometers and vibration sensors in vehicles to monitor for crashes or rollovers. Upon detecting a severe accident, the system promptly sends an alert message via a GSM module, containing GPS coordinates, to relevant authorities or vehicle owners. This facilitates rapid emergency response, particularly in isolated areas, potentially saving lives by minimizing communication delays.

Abdul khadir shehu bari, Abubakar Falalu, Aawal Umar and Mansur Gamble [15] The proposed method of the system involves using a vibration sensor to detect car accidents with location information provided by GPS modules. When an accident is detected, the system notifies the nearest emergency services and medical centers via GSM and GPS modules. The system also includes a "HELP ME" button for emergency situations other than accidents. The location data is transmitted to the emergency center via the GSM module and sent as a text message to everyone on the emergency list.

M. Phani Durga, E. P. L. Jaya, and K. Madhavi [16] propose a method for accident detection utilizing an accelerometer to identify the tilting and crashing of the vehicle. Subsequently, the system sends the GPS location of the accident scene to designated security, medical, and family contacts. It dispatches an SMS containing Minimum Set of Data (MSD) information, comprising geographical coordinates and a link that opens directly in Google Maps. Additionally, within 3 seconds of sending the SMS, the system initiates a voice call to the designated contacts.

Rajat Amat and Sunil Mallick [17] propose a system designed to sound an alarm when an accident occurs, employing Arduino, sensors, GSM, and GPS technology. The system comprises a single-board embedded system equipped with GSM and GPS, alongside a microcontroller. Through the utilization of GSM and GPS technologies, the system tracks the vehicle and provides real-time information about ongoing trips. Monitoring data from the accelerometer and vibration sensor enables the system to identify severe accidents. Upon detection, it sends an alert message containing latitude and longitude data, obtained from the GPS module, to the police control room, rescue teams, or the vehicle owner via the GSM module.

Mohd Akram Khan, Nirbhay Pandey, Ajay Kumar, and Shadab Ali [18] introduce a method for an Accident Detection and Alert System utilizing an Android Application. This method involves employing machine learning algorithms to analyze data from GPS and accelerometer sensors for accident detection. Subsequently, the system promptly notifies emergency services and loved ones.

M. Kiran Kumar, Aniruddha Balbudhe, and CH Sai Karthikeya [19] put forth a model leveraging IoT-based technology, employing sensors, microcontrollers, and cloud infrastructures to detect accidents involving two-wheeled motorcycles. This system has the capability to recognize and report accidents, issue warning notifications to the rider, and inform emergency services and predetermined contacts of the rider's location and condition. The proposed system entails a multifaceted process of sensor selection, software development, and integration of all components into an operational system.

P. Josephinshermila, S. Sharon Priya, K. Malarvizhi, and Ramakrishna Hegde [20] propose the utilization of an Automotive Smart Black-Box based Monitoring System to detect and alert emergency services about vehicle accidents. This system employs sensors to monitor vehicle parameters and stores data both on a digital memory card and in the cloud. Actively detecting accidents, the proposed model ensures real-time updates of information. To enhance accuracy, the system utilizes efficient communication techniques like Zigbee, GSM, and GPS.

III PROPOSED SYSTEM

A strategy to bolster road safety and emergency medical response harnesses IoT (Internet of Things) technology. This holistic solution aims to meet the urgent demand for swift reactions to road accidents, where timing can be critical.Fig.1 shows the architecture of the system. Vehicles are equipped with IoT devices containing advanced sensors such as accelerometers and gyroscopes, continuously collecting data on vehicle movements and conditions. This data is instantly transmitted to a centralized server, forming the foundation of the system.

At the core of this system lies an innovative accident detection algorithm. This algorithm analyzes incoming data to detect patterns indicative of road accidents, considering factors like abrupt changes in vehicle acceleration, orientation, and other parameters. Upon detecting an accident, the system activates an immediate hospital alert mechanism. Integrated with a comprehensive database of local hospitals, including their locations and capacities, the hospital alert system promptly dispatches accident alerts to these institutions in real-time. Hospitals receive alerts via a dedicated web application, enabling them to swiftly assess the situation and respond accordingly. They can either accept or decline the request for assistance based on their available resources and capabilities.

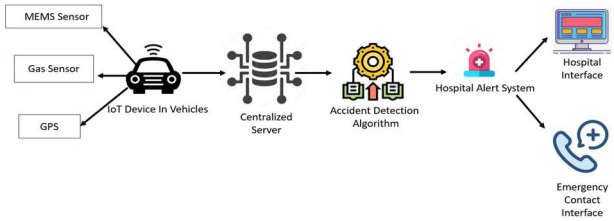

Fig 1. Architecture Diagram of proposed System

If the preferred hospital cannot accommodate the injured individuals, the system utilizes an intelligent fallback

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mechanism. This mechanism identifies the closest available healthcare facility, guaranteeing that accident victims receive timely medical attention. A critical aspect of this strategy is its ability to adapt to varying scenarios and ensure effective emergency responses. Additionally, continuous monitoring and optimization of the system's performance are essential to maintain its reliability and efficiency. By leveraging IoT technology and integrating with existing healthcare infrastructure, this solution holds promise in significantly reducing response times to accidents and improving overall road safety. With its robust architecture and intelligent features, it underscores the importance of leveraging technological advancements to save lives and enhance emergency medical services.

IV METHODOLOGIES USED

1.Installation of IoT Devices:

The installation of IoT devices within vehicles involves deploying sophisticated sensors such as MEMS (Micro-Electro-Mechanical Systems) in Fig.2 and gas sensors in Fig.3. These sensors are strategically placed to continuously gather detailed data on various aspects of vehicle dynamics, including acceleration, orientation changes, and gas levels. By capturing this intricate data in real-time, the IoT devices lay the foundation for effective accident detection and emergency response systems.

Fig 2. MEMS Sensor

Fig 3. Gas Sensor

2.Establishment of Data Transmission Infrastructure:

A robust data transmission infrastructure is established to facilitate the seamless transmission of real-time data from vehicles to a centralized server. This infrastructure ensures that accident-related data is swiftly relayed to the centralized server, enabling timely analysis and response. By establishing reliable communication channels, the system ensures the accessibility and integrity of critical information during emergency situations..

3.Configuration of Centralized Server:

The centralized server serves as the repository for all incoming data, playing a pivotal role in the system architecture. Configured to handle large volumes of data, the server hosts essential components necessary for seamless data processing and analysis. By centralizing data management, the server facilitates efficient data retrieval, analysis, and dissemination, ensuring the smooth operation of the accident detection and response system.

4.Development of Accident Detection Algorithm:

An advanced accident detection algorithm is developed to analyze incoming sensor data and identify patterns indicative of accidents in real-time. Leveraging machine learning techniques, particularly the Random Forest model, the algorithm achieves an impressive accuracy rate of 98%. By detecting accidents swiftly and accurately, the algorithm forms the core functionality of the system, enabling prompt emergency responses and potentially saving lives.

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Accuracy: 0.98
Precision: 0.9186046511627907
Recall:0.9
F1 Score: 0.90909090909090917
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Fig 4. Performance Metrics and Evaluation

5.Creation of Hospital Database:

A comprehensive database containing information on local hospitals is compiled, including their geographical locations and available resources. This database serves as a crucial resource for informed decision-making during emergency situations, allowing the system to efficiently dispatch accident alerts to nearby hospitals based on their capacities and capabilities.

6.Implementation of Alert Trigger Mechanism:

A robust mechanism is designed to promptly trigger alerts to hospitals upon the detection of potential accidents by the algorithm. Quick notification is essential for facilitating efficient emergency medical response, and this mechanism ensures that hospitals receive timely alerts, enabling them to assess the situation and respond promptly to assistance requests

7.Development of Hospital Response Interface:

A dedicated web application is crafted for hospitals to receive and respond to accident alerts. Fig.5 shows the interface empowers hospitals to assess their capacity and promptly respond to assistance requests. With user-friendly features and real-time updates, the interface facilitates seamless communication between the accident detection system and healthcare providers, optimizing emergency medical response efforts.

Fig 5. Hospital Response Interface

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8.Integration of Fallback Mechanism:

An intelligent fallback mechanism is incorporated into the system to reroute alerts to the nearest healthcare facility if the primary hospital cannot accommodate the injured individuals. This ensures that accident victims receive timely medical attention, even if the initially targeted hospital is unavailable. By dynamically adapting to changing circumstances, the system maximizes the chances of swift emergency responses and effective patient care.

9.Inclusion of Alert Cancellation System:

An alert cancellation system is integrated into the system, enabling hospitals or authorized personnel to cancel alerts if further assessment reveals no need for medical assistance. This feature prevents unnecessary deployment of medical resources and ensures efficient allocation of healthcare resources during emergency situations. By allowing for timely cancellation of alerts, the system optimizes resource utilization and enhances overall system efficiency..

V RESULTS

The exceptional accuracy rate of 98% achieved by our accident detection algorithm instills confidence in the reliability of our system, providing assurance that genuine accidents will be promptly identified and addressed. This reliability is essential for optimizing emergency response efforts and ensuring the efficient allocation of resources during critical situations on the road.Moreover, the seamless integration of IoT technology in our system not only enhances accident detection capabilities but also facilitates swift communication with relevant stakeholders, including emergency services and healthcare facilities. This interconnectedness enables coordinated responses to accidents, reducing response times and improving overall outcomes for accident victims.Furthermore, the exemplary performance highlighted in Fig.6 and Fig.7 underscores the transformative potential of our system in revolutionizing emergency medical services and enhancing road safety standards. By harnessing IoT technology for instantaneous accident detection and hospital alerting, we aim to establish a new paradigm in emergency response systems.

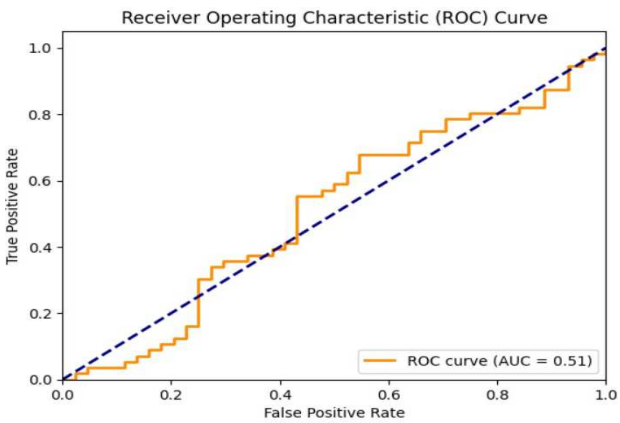

Fig 6. Receiver Operating Characteristic (ROC) Curve

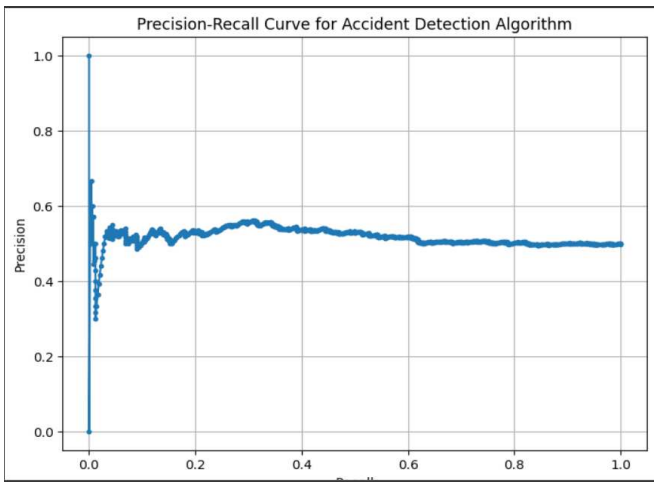

Fig 7. Precision-Recall Curve

VI CONCLUSION

In this study, our efforts to incorporate IoT technology into accident detection and hospital alert systems unveil a promising path for improving emergency healthcare services. Through adept utilization of IoT, we've developed a solution that markedly diminishes response times during accidents while streamlining resource allocation for vital medical aid. In a realm where swift and effective responses are paramount, this system carries the potential to rescue lives and elevate the overall standard of emergency healthcare services. It highlights the significance of real-time monitoring and communication within the healthcare sphere, laying the groundwork for a more effective, agile, and life-preserving approach to managing accidents and emergencies.

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